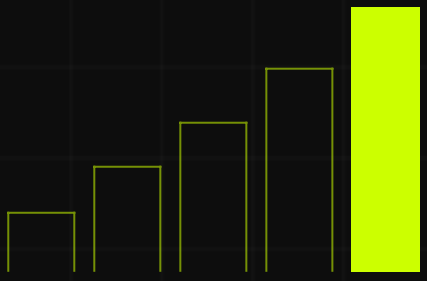
VISIONSEEK

VISIONSEEK LAB · DOCTRINE PAPER

The Localization Doctrine

From using technology to owning it:
Korea is the proof that the Arab world can

VisionSeek Lab · Doctrine Paper · Insights



October 2026

Feeds the Arab Technological Capability Index and the Incheon Dialogue (April 2027)

ABOUT

About this paper

A doctrine paper from VisionSeek Lab in the Insights series. It reads South Korea's experience in seven technologies not as a subject but as proof that moving from buying technology to owning it is possible within one working lifetime.

It feeds the Arab Technological Capability Index and the Incheon Dialogue (April 2027).

How to read the source markers

[4] Confirmed by source; the number refers to the source list at the end.
[Press 6] Secondary press source.
◇ Inference VisionSeek analysis.
◦ Gap Outside the scope of this edition.

> This edition rests on 31 sources: 28 primary, two official records and one press source. All figures are Korean or international.

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What will we own when this contract ends?

The question this paper proposes for every coming Arab technology deal

THIS IS OUR MOMENT

This is our moment

Korea is not the subject of this report. It is the proof

A country that starts as a buyer can become an owner within one working lifetime.

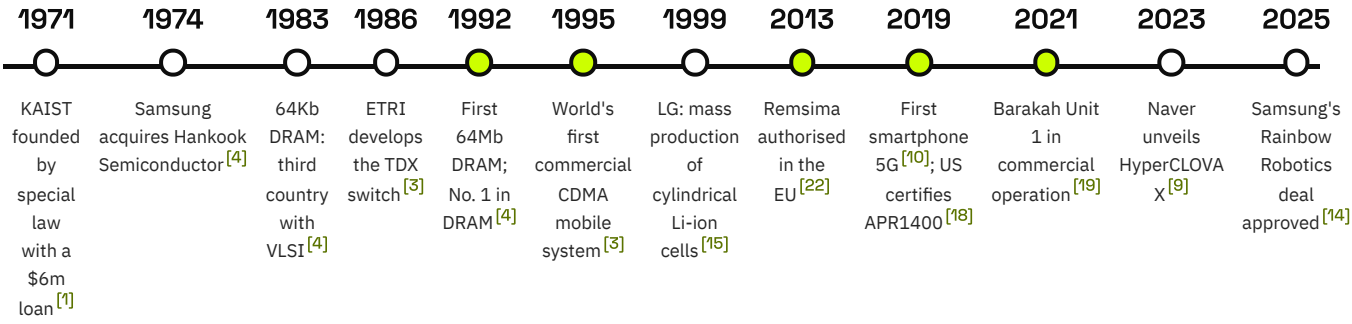
For two generations the Arab world has been an excellent *buyer* of technology. We bought the best networks, the best power plants and the best devices, and every purchase renewed our dependence. The question was never whether we could afford technology. It was whether we would ever *own* it.

Korea answers that question. In 1971 it founded its first science and technology university under a special law, with a USAID loan of six million dollars[1]. In 1983 Samsung developed a 64Kb DRAM, making Korea the third country in the world with advanced VLSI technology[4]. By 1992 Samsung held the world's largest DRAM market share[4]. Korea is not the subject of this report. It is the **proof** that a country that starts as a buyer can become an owner within one working lifetime.

For the first time, several of the conditions behind Korea's leap are present in our region at once. This is the time we have waited for. The danger is that we spend it buying again.

Exhibit 1 **From buyer to owner: Korean milestones documented in this paper**

Selected dates from the text, in order; highlighted points are "ownership" milestones (a world first, market leadership, or certification and operation abroad)



Sources: [1] [3] [4] [9] [10] [14] [15] [18] [19] [22]; see source list. The "ownership milestone" classification is VisionSeek analysis ◇ Inference.

EXECUTIVE SUMMARY

Executive summary: five conclusions

1 Ownership is a sequence, not a purchase.

Korea moved from import to assembly, then absorption, then its own design, then setting the standard. Foreign technology was the *first* step, never the last ◇ Inference.

2 The decisive asset is the "absorption institution."

KAIST was created by special law with an explicit national mission[1], and ETRI carried Korea from telephone switches to CDMA to 6G[3]. Arab states have invested in purchases. Few have built an institution that absorbs what they buy ◇ Inference.

3 Export is what proves ownership.

The Korean APR1400 reactor counted as owned once it was certified abroad[18] and running in the UAE[19]. Even then, an intellectual-property dispute over its origins had to be settled[20] ◇ Inference (lesson).

4 Settle rights at the door.

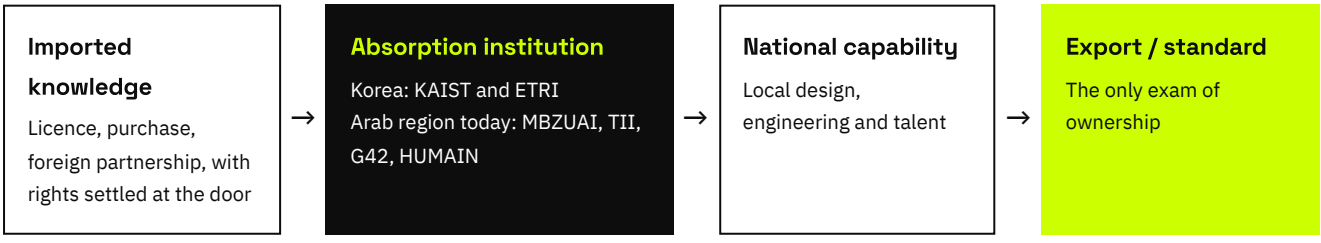
Korea's battery leaders ended up before a US trade tribunal over trade secrets[17], and its reactor exporter had to settle with Westinghouse[20]. IP left unresolved when technology is transferred comes back when it is exported ◇ Inference.

5 Integrity is infrastructure.

The landmark 2005 Korean stem-cell paper in *Science* was retracted[24]. Faked capability destroys trust faster than real capability builds it, so an Arab capability index must measure ownership that has been verified, not announced ◇ Inference.

Exhibit 2 The absorption institution: the loop that turns purchase into ownership

Conceptual model drawn from conclusions 2, 3 and 4; not data



Source: VisionSeek analysis ◇ Inference; institution names[1, 3, 28–31].

WHY NOW

Why now: what has actually changed

1 **National visions with diversification as an explicit goal**

Saudi Arabia's Cabinet approved Vision 2030 on 25 April 2016[25], and the document commits to diversifying the economy[26]. Egypt launched Vision 2030 in 2016 and has since updated it[27]. Visions built on purchases stall; visions built on absorption compound ◇ Inference.

2 **Arab AI institutions now exist**

The UAE made AI a national priority in 2017, appointed the world's first Minister of State for AI, and in 2019 announced MBZUAI, described as the first graduate-level, research-based AI university, whose first class started in September 2020[28]. TII is part of Abu Dhabi's Advanced Technology Research Council[29]. G42 describes itself as "born in Abu Dhabi, operating globally"[30]. In 2025 Saudi Arabia's PIF launched HUMAIN to work across the AI value chain, including a multimodal Arabic large language model[31]. These are the first Arab counterparts to Korea's KAIST/ETRI layer ◇ Inference.

3 **The AI window is open**

AI capability is still forming, and entry into applied, sector-specific and Arabic-language models costs far less than leading-edge chip fabs ◇ Inference.

4 **Governments are redrawing supply chains**

The US CHIPS and Science Act provides $52.7 billion and a 25% investment tax credit, and the White House says the US relies on East Asia for 75% of chip production[8]. Korea expanded its own semiconductor tax support in 2023[7]. When the leaders subsidize relocation, partners look for new, stable sites rich in energy and capital, and the Arab region is one of them ◇ Inference.

5 **Korea is looking at us**

Korea's 6G strategy names the Middle East as a priority region for its digital exports[11]. A partner that wants our market is a partner that can be asked for absorption terms ◇ Inference.

6 **A Korean ownership proof already stands on Arab soil**

Barakah has four APR-1400 units, construction began in 2012, and Unit 1 entered commercial operation in April 2021[19].

FIGURES FROM THE TEXT

$52.7bn
US CHIPS and Science Act funding[8]

25%
Investment tax credit in the same Act[8]

75%
Share of chip production for which the US relies on East Asia, per the White House[8]

4 × APR-1400
Korean units at Barakah; Unit 1 commercial in April 2021[19]

> **Visions built on purchases stall; visions built on absorption compound.**

01

PART ONE · SEVEN TECHNOLOGIES

Seven technologies, seven lessons

For each technology: what Korea did, the mechanism, what failed, what applies to the Arab world and what does not, and the Arab opportunity open now.

1	2	3	4	5	6	7
Semiconductors	Artificial intelligence	Cloud & data	5G / 6G	Robotics	Clean energy & batteries	Biotech & health tech

OVERVIEW

Map of the seven technologies

In every technology Korea entered through a narrow door, then climbed. The Arab opportunity is not to repeat the same path but to choose the beachhead that fits our own problems ◇ Inference.

Exhibit 3 **Seven technologies: Korea's entry point, the Arab opportunity, and documentation level**

Summary of the sections that follow

Technology	Korea's entry point	Arab opportunity open now ◇ Inference	Documentation
Semiconductors	Acquisition of Hankook Semiconductor (1974) [4], then DRAM technology from Micron under contract [Press 6]	Packaging and testing, chip design for energy and water, power semiconductors for the grid	▮ Documented; market shares: gap
AI	A national platform: Naver's HyperCLOVA X (2023) [9]	Arabic models for health, education, public services, law and heritage	▮ Documented; failures: gap
Cloud & data	Cloud Security Assurance Program (CSAP) for public-sector entry [12]	Sovereign-cloud certification requiring local operations and engineering teams	▮ Documented; mechanism: inference
5G / 6G	A state lab (ETRI) co-developing with industry: TDX, then CDMA, then LTE-A [3]	Contributing to 6G standards and testbeds	▮ Documented; failures: gap
Robotics	KAIST humanoid researchers found Rainbow Robotics (2011) [13]	Robots for heat, desert logistics, construction, hospitals and Hajj and Umrah crowd services	▮ Documented; failures: gap
Energy & batteries	LG lithium-ion research [15–16]; APR1400 reactor [18]	Grid-scale storage for solar; battery-materials processing	■ Documented
Biotech	Manufacturing to Western standards: Celltrion and Samsung Biologics [21–23]	A regional base for biosimilars and vaccine fill-finish for Arab and African markets	■ Documented

Sources: as marked in each cell. "Documentation" is an editorial assessment of source coverage, not a quantitative measure.

> **Choose one narrow beachhead. Korea won memory, not all chips.**
>
> Principle 3 of the Localization Doctrine

1 Semiconductors

What Korea did

Samsung entered the field by acquiring Hankook Semiconductor in 1974[4]. Its founder made the "Tokyo declaration" on 8 February 1983[5]. That same year it developed the 64Kb DRAM[4], having taken basic DRAM technology from Micron under contract, and its trainees had only limited access during the transfer[Press 6]. In 1992 it developed the industry's first 64Mb DRAM and reached the No. 1 DRAM market share[4]. ETRI, the state research institute, developed a 4M DRAM in 1989[3].

Mechanism

License, then absorb by force of necessity when the licensor holds back, then leap ahead in *one* narrow segment (memory) where manufacturing discipline beats design breadth ◇ Inference. A state research institute ran in parallel with the firms[3] ◇ Inference (role).

What failed

Market shares are outside the scope of this edition ◦ Gap. The risks of concentrating on memory, such as price cycles, are an inference ◇ Inference.

Arab transfer

What applies: choose one segment, and write absorption clauses into every licence. What does not: copying a leading-edge fab, because Korea's window of the 1980s is closed ◇ Inference.

Arab opportunity open now

Packaging and testing, chip design for our own sectors (energy, water), and power semiconductors for the grid ◇ Inference; needs a feasibility study.

Exhibit 4 License → absorb → leap: Korea's memory path in four milestones

Documented milestones from the text; phase labels are VisionSeek analysis

Entry	Licence	Absorption	Leap
1974	**1983**	**1989**	**1992**
Samsung acquires Hankook Semiconductor[4]	"Tokyo declaration"[5]; DRAM technology from Micron under contract, limited access[Press 6]; 64Kb DRAM[4]	State institute ETRI develops 4M DRAM in parallel with the firms[3]	Industry's first 64Mb DRAM; No. 1 DRAM market share[4]

Sources: [3] [4] [5]; [6] is a secondary press source and the only one for the Micron transfer details. Phase labels ◇ Inference.

Absorption by force of necessity when the licensor holds back.

2 Artificial intelligence

What Korea did	On 24 August 2023 Naver unveiled HyperCLOVA X, its own generative AI, together with the CLOVA X assistant and CUE: search[9]. It is a domestic platform built on its own data and infrastructure.
Mechanism	A national platform company owns the language, search and data layer ◇ Inference.
What failed	Outside the scope of this edition ◦ Gap.
Arab transfer	What applies: language sovereignty, since Arabic is a market large enough to justify models we own ◇ Inference. What does not: competing on frontier compute alone ◇ Inference.
Arab opportunity open now	Arabic models for health, education, public services, law and heritage. Institutions that can carry this already exist[28, 29, 31] ◇ Inference (opportunity).

> **Arabic is a market large enough for models we own.**

3 Cloud & data

What Korea did	Korea runs a Cloud Security Assurance Program (CSAP) for cloud providers serving the public sector. MSIT and the National Intelligence Service have announced that it will be merged into a single NIS-led verification system, fully in place in the second half of 2027[12].
Mechanism	The state sets the rules for entering the public market. Sovereignty is exercised through certification, not ownership of every data centre ◇ Inference.
Arab transfer	A data centre on our soil is *use*. Owning the operating software, the data standards and the trained engineers is *ownership* ◇ Inference.
Arab opportunity open now	Sovereign-cloud certification that requires local operations and engineering teams, so that every data centre becomes a site of absorption ◇ Inference.

4 5G / 6G

What Korea did

ETRI developed the TDX switch in 1986, commercialized the world's first CDMA mobile system in 1995, and went on to WiBro in 2004 and LTE-Advanced in 2010[3]. On 3 April 2019 Korea achieved the world's first smartphone-based commercial 5G service[10]. In February 2023 MSIT launched the K-Network 2030 strategy. It runs a preliminary feasibility study for KRW 625.3 billion of 6G R&D, plans a Pre-6G Vision Fest in 2026, and targets a 5G patent share above 30%, up from 25.9% according to IPlytics as cited by MSIT[11].

Mechanism

A state lab co-develops with industry, the national operators act as first buyers, and the state then sets the standards agenda[3, 11] ◇ Inference (mechanism).

What failed

Outside the scope of this edition ◦ Gap.

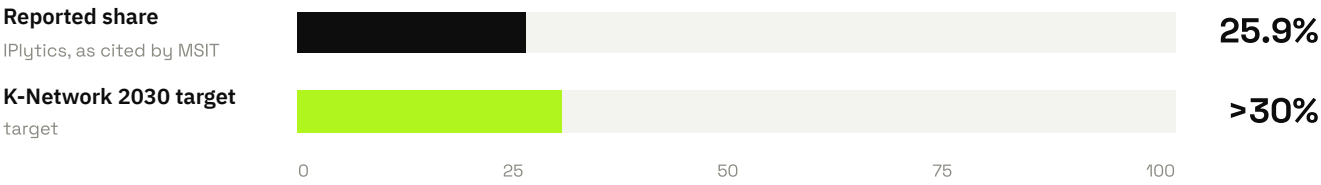
Arab transfer

What applies: the state as a *demanding first customer*, with local R&D obligations. What does not: building a full national equipment vendor ◇ Inference.

Arab opportunity open now

6G is still before standardization, so now is the time to contribute to standards and testbeds rather than buy them later. Korea itself is seeking Middle East partners[11] ◇ Inference.

Exhibit 5 From participant to standard-setter: Korea's 5G patent share and target



Source: [11] Korea MSIT, K-Network 2030 strategy (February 2023), citing IPlytics. The "above 30%" target is drawn at its lower bound.

5 Robotics

What Korea did

Rainbow Robotics was founded in 2011 by researchers from KAIST's Humanoid Robot Research Center, the team behind Hubo, Korea's first two-legged walking robot. Samsung bought 14.7% in 2023 for KRW 86.8 billion and moved to 35%[13]. Korea's Fair Trade Commission approved the deal on 5 March 2025[14].

Mechanism

A university lab becomes a spin-out, which is then acquired by a national champion, closing the capital loop ◇ Inference.

What failed

Outside the scope of this edition ◦ Gap.

Arab opportunity open now

Robots for our own hard problems, such as heat, desert logistics, construction, hospitals, and crowd services during Hajj and Umrah ◇ Inference.

6 Clean energy & batteries

What Korea did

LG Energy Solution dates its lithium-ion research to 1992 and the mass production of cylindrical lithium-ion cells to 1999[15]. LG Chem's own archive dates the start of battery research to 1995[16], so the two official sources differ. In nuclear, the US NRC certified the APR1400 design with effect from 19 September 2019[18], and four units were built at Barakah in the UAE[19].

What failed

SK and LG fought a battery trade-secret dispute before the US International Trade Commission, which they settled in April 2021[17]. Westinghouse and KEPCO/KHNP settled an IP dispute in January 2025 on confidential terms[20]. Lesson: **IP left unresolved at transfer returns at export** ◇ Inference.

Arab transfer

Barakah shows that Arab states can host ownership. The next step is to require Arab engineering roles in operation and design ◇ Inference.

Arab opportunity open now

Grid-scale storage for solar, and processing battery materials where the region has energy and capital ◇ Inference.

> **IP left unresolved at transfer returns at export.**

7 Biotech & health tech

What Korea did

The EMA recommended Remsima (Celltrion), together with Inflectra, as the first two monoclonal-antibody biosimilars[21], and Remsima was authorised across the EU on 10 September 2013[22]. Samsung Biologics opened its Plant 5 in April 2025, bringing total capacity to 784,000 litres[23].

Mechanism

Enter through *manufacturing* to Western regulatory standards, then move up toward innovation ◇ Inference.

What failed

The 2005 *Science* stem-cell paper was retracted[24].

Arab opportunity open now

A regional base for biosimilars and vaccine fill-finish serving Arab and African markets, with ethical governance of genomic data within the bounds of Sharia and medical ethics ◇ Inference.

02

PART TWO · THE DOCTRINE

The Localization Doctrine

We have capital, energy and young people. What we have lacked is a doctrine: to turn every purchase into a lesson, and every lesson into ownership.

1	2	3	4	5
Use	Assemble	Adapt	Design	Export / Standard

EIGHT PRINCIPLES

The Localization Doctrine: eight principles ◇Inference

Eight principles drawn from the seven technologies, usable as a checklist for any Arab technology deal or partnership.

Exhibit 6 **The eight principles of the Localization Doctrine**

Conceptual framework from VisionSeek analysis; small lines point to the Korean evidence in the paper's sections

1

Buy to learn, never buy to keep.

Every contract names what we will absorb, who absorbs it, and by when.

2

Build the absorbing institution before the deal.

KAIST by special law[1]; ETRI[3]

3

Choose one narrow beachhead.

Korea won memory, not all chips.

4

The state is first customer and standard-setter, not operator.

Operators as first buyers; the state leads the standards agenda[3, 11]

5

Keep the core (R&D, pilot lines, quality control) at home in every joint venture.

A VisionSeek proposal, not a description of Korea.

6

Settle IP at the door.

SK–LG settlement[17]; Westinghouse settlement[20]

7

Export is the exam.

APR1400 certified in the US[18] and running at Barakah[19]

8

Verify, or it didn't happen.

Stem-cell paper retraction (2005)[24]

Source: VisionSeek analysis ◇Inference; evidence linked to its sources. Linking evidence to principle is editorial analysis.

Buy to learn, never buy to keep.

Every contract names what we will absorb, who absorbs it, and by when.

THE OWNERSHIP LADDER

The ownership ladder: from use to setting the standard

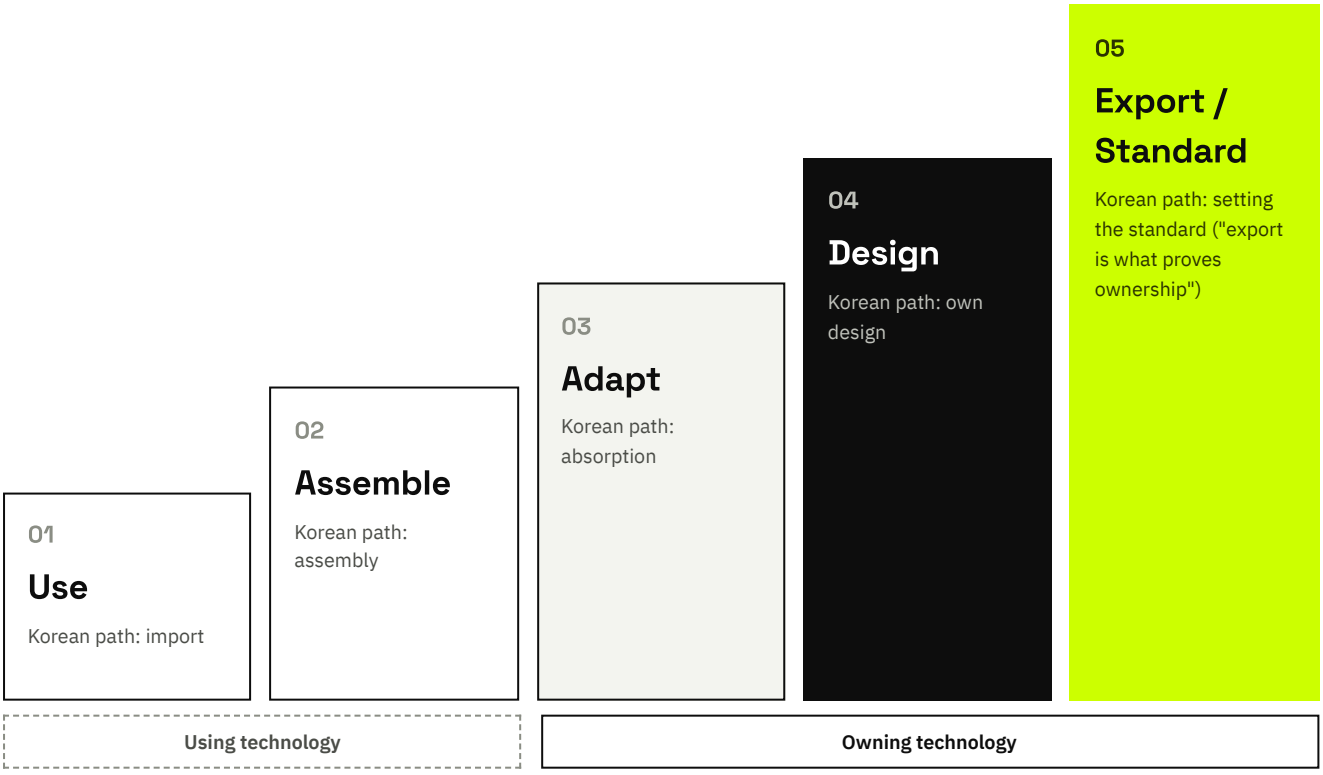
For the Arab Technological Capability Index we propose a five-step ladder for each country and each technology ◇ Inference.

It mirrors the sequence Korea followed: from import to assembly, then absorption, then its own design, then setting the standard ◇ Inference.

Ownership is a sequence, not a purchase. Foreign technology was always the first step, never the last. An Arab index must measure ownership that has been verified, not announced.

Exhibit 7 **The five-step ownership ladder, proposed for the Arab Technological Capability Index**

Conceptual model; step heights are illustrative and do not represent measured values



Source: VisionSeek analysis ◇ Inference. The use/own split is illustrative and follows the paper's thesis that export is the exam of ownership.

Export is what proves ownership.

The Korean APR1400 reactor counted as owned once it was certified abroad [18] and running in the UAE [19].

A CALL TO ARAB DECISION-MAKERS

A call to Arab decision-makers: this is our moment.

Korea started its first research university with a six-million-dollar loan[1]. We have capital, energy, young people and, for the first time, our own AI institutions[28–31]. What we have lacked is a doctrine: the discipline to turn every purchase into a lesson and every lesson into ownership.

ONE QUESTION FOR YOUR NEXT TECHNOLOGY DEAL

In your next technology deal, ask one question: what will we own when this contract ends?

If there is no answer, the deal is not finished.

Let the Incheon Dialogue in April 2027 be where we start, and the moment the Arab world stopped buying the future and began building it.

SOURCES & METHODOLOGY

Sources and methodology

[n] Confirmed by source
The number refers to the list below.

[Press n]
Secondary press source, used only where marked.

◇ Inference
VisionSeek analysis built on the documented cases.

◦ Gap
Outside the scope of this edition.

All sources were opened and read on 24 October 2026. 31 sources: 28 primary (government, regulator, company or institution), two official records and one press source. The paper cites only what each page actually says.

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Scope and sources

- This edition covers Korea's experience in seven technologies, from semiconductors to biotech, and draws from it eight principles and a five-step ownership ladder.
- It rests on 31 sources: 28 primary, two official records and one press source [6] for the Micron transfer details. Every fact is tagged with its source and every analysis is tagged "Inference".
- All figures are Korean or international; Arab measurement is the domain of the Arab Technological Capability Index.
- The next edition adds: Korea's AI, telecoms and robotics shortfalls, semiconductor market shares, Korean semiconductor tax-credit rates, Korean robotics policy, and comparative data from the OECD, World Bank, IEA and WIPO.

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From using technology to owning it.

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